Fine-Tuning BART for English-Turkish Machine Translation

Alperen Balım
Department of Computer Engineering
Izmir Katip Celebi University
210401021

Serdar Can
Department of Computer Engineering
Izmir Katip Celebi University
220401096

Efekaan Ketenci
Department of Computer Engineering
Izmir Katip Celebi University
220401060

***Abstract*—Transformer-based architectures have significantly advanced the state of Natural Language Processing, particularly in Sequence-to-Sequence tasks such as Machine Translation [1]. This study investigates the fine-tuning process of the pre-trained BART-base model for English-to-Turkish translation. Utilizing a 50,000-sentence subset of the OPUS-100 dataset [2], a comprehensive training pipeline was established to adapt the model to the target language pair. The model was trained over three epochs and evaluated using the SacreBLEU metric [3]. Results indicate a stable decrease in training loss to 1.8 and a final validation BLEU score of 9.9, demonstrating the model's capacity to learn bilingual syntactic mappings.**

***Index Terms*—Natural Language Processing, Machine Translation, BART, Fine-Tuning, Seq2Seq**

I. Introduction

Neural Machine Translation (NMT) relies heavily on Sequence-to-Sequence (Seq2Seq) models to map text from a source language to a target language. The Bidirectional and Auto-Regressive Transformers (BART) architecture integrates a bidirectional encoder with a left-to-right autoregressive decoder, making it highly suitable for generative tasks like translation.

This paper details the process of fine-tuning the *facebook/bart-base* model specifically for the English-Turkish language pair. By defining a clear preprocessing protocol, hyperparameter configuration, and algorithmic training structure, we demonstrate how pre-trained models can be systematically adapted to distinct linguistic tasks using parallel corpora.

II. Literature Review

Neural Machine Translation (NMT) has achieved remarkable success across high-resource language pairs, yet its application to morphologically rich and low-resource languages such as Turkish remains challenging [4]. Turkish is characterized as a highly agglutinative language with a complex morphological structure, which frequently leads to data sparsity issues during model training [4], [5]. Furthermore, the grammatical structure and word order differ significantly between the two languages; Turkish follows a Subject-Object-Verb (SOV) order, contrasting with the Subject-Verb-Object (SVO) order of English [6].

To address the challenges posed by morphologically rich languages, previous studies have explored various architectures beyond traditional recurrent neural networks. The paradigm shift initiated by Vaswani et al. [9] introduced the Transformer architecture, completely replacing recurrence with self-attention mechanisms, thereby allowing models to capture long-range dependencies more effectively. Following this, handling the Out-Of-Vocabulary (OOV) problem in agglutinative languages became a primary focus. Sennrich et al. [10] demonstrated that using subword units, specifically Byte-Pair Encoding (BPE), significantly improves the translation of rare and unseen words by breaking them down into frequent character sequences.

Some approaches have also utilized morphological word embeddings or integrated external linguistic features, such as Part-of-Speech (POS) tags, to enrich the representation of words and mitigate data sparsity [5], [7]. Recent research has demonstrated the effectiveness of transfer learning and pre-trained language models in improving low-resource translation for the English-Turkish language pair [7], [8]. Building upon these foundations, our work leverages the pre-trained BART architecture—utilizing BPE tokenization and bidirectional encoding—to efficiently map English semantic representations into Turkish, overcoming the structural disparities highlighted in the literature.

III. Methodology

A. Dataset and Preprocessing

The translation data is sourced from the Helsinki-NLP/opus-100 dataset [2], an English-centric multilingual corpus covering 100 languages. A random subset of 50,000 English-Turkish sentence pairs was extracted and partitioned using a 90/10 split (45,000 samples for training and 5,000 for validation).

Text tokenization was performed using the *BartTokenizer*, which inherently relies on Byte-Pair Encoding (BPE) [10]. This subword tokenization strategy is particularly advantageous for the Turkish language, as it dynamically splits complex agglutinative words into recognizable sub-tokens, thereby preserving the semantic roots and morphological suffixes without inflating the vocabulary size.

Source and target sequences were truncated to a maximum length of 128 tokens. This threshold was empirically selected to cover the majority of sentence lengths in the OPUS-100 subset while optimizing computational memory during the attention matrix calculations. Padding was applied dynamically via a Data Collator, and target labels corresponding to padding

tokens were masked (assigned an ID of -100) to exclude them from the cross-entropy loss computation, ensuring the model focuses exclusively on valid syntactic mappings.

B. Model Architecture

The *BART-base* architecture [1] is employed to provide a robust baseline for capturing the morphological complexities of Turkish and mapping them to English semantics. The core structural parameters, extracted directly from the model's configuration file, are detailed in Table I.

TABLE I
BART-BASE ARCHITECTURAL CONFIGURATIONS

Hyperparameter	Value
Hidden Dimension (d_{model})	768
Encoder Layers	6
Decoder Layers	6
Attention Heads	12
Feed-Forward Network (FFN) Dimension	3072
Vocabulary Size	50,265
Maximum Position Embeddings	1024
Activation Function	GELU
Dropout (Activation / Attention)	0.1

Additionally, the overarching pipeline detailing the data flow from text tokenization through the encoder-decoder attention mechanism to the final target generation is illustrated in the architectural schema in Fig. 1.

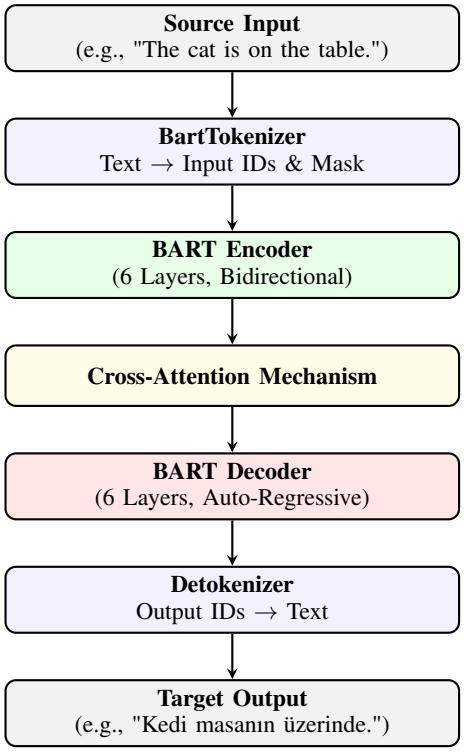

Fig. 1. High-level schematic of the Sequence-to-Sequence fine-tuning pipeline for English-Turkish machine translation.

C. Training Algorithm

The fine-tuning phase was managed by standardizing learning parameters and iterative optimization steps. The learning rate was set to 5×10^{-5} with a weight decay of 0.01. The training sequence is formally defined in Algorithm 1.

Algorithm 1 BART Fine-Tuning Process for Machine Translation

Require: OPUS-100 en-tr dataset $\mathcal{D}$, Pre-trained BART model $\mathcal{M}$, Tokenizer $\mathcal{T}$

$\mathcal{D}_{subset} \leftarrow$ Select 50,000 sentence pairs from $\mathcal{D}$
$\mathcal{D}_{train}, \mathcal{D}_{val} \leftarrow$ Split $\mathcal{D}_{subset}$ (Ratio: 90% / 10%)
$\mathcal{D}_{train}^{tok} \leftarrow$ Tokenize($\mathcal{D}_{train}, \mathcal{T}$, max_length = 128)
$\mathcal{D}_{val}^{tok} \leftarrow$ Tokenize($\mathcal{D}_{val}, \mathcal{T}$, max_length = 128)
Initialize DataCollatorForSeq2Seq with $\mathcal{T}$ and padding configuration
Define Training Arguments (Epochs=3, Batch Size=2, LR=5×10^{-5})
Initialize Trainer with $\mathcal{M}, \mathcal{D}_{train}^{tok}, \mathcal{D}_{val}^{tok}$, DataCollator, and SacreBLEU
for each epoch $\in \{1, 2, 3\}$ **do**
  for each batch $\in \mathcal{D}_{train}^{tok}$ **do**
    Compute cross-entropy loss
    Backpropagate gradients
    Update model weights $\mathcal{M}$
  end for
  Compute BLEU score on $\mathcal{D}_{val}^{tok}$
  Save model checkpoint
end for
return Fine-tuned model weights

IV. Experimental Results and Discussion

Translation quality was evaluated using the SacreBLEU metric [3]. During generation, a beam search algorithm (beam width = 4) and a tri-gram repetition penalty were applied to ensure fluency and reduce repetitive outputs.

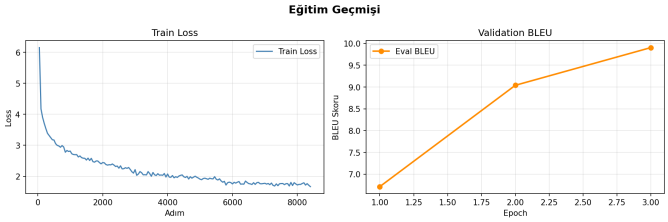

Fig. 2. Training history displaying the Train Loss over steps (left) and Validation BLEU score over epochs (right).

The quantitative performance of the final model (Epoch 3) on the validation set is summarized in Table II. In addition to the standard BLEU metric, chrF was included to better account for character-level n-gram matches, which is critical for an agglutinative language like Turkish. Furthermore, the Translation Edit Rate (TER) is reported to indicate the post-editing effort required.

TABLE II
EVALUATION METRICS ON THE VALIDATION SET (EPOCH 3)

Model	**BLEU** ↑	**chrF** ↑	**TER** ↓
BART-base	9.90	30.11	83.21

As depicted in Fig. 2, the training loss decreased steadily from an initial value above 6.0, stabilizing around 1.8 by the end of the optimization steps. Concurrently, the validation BLEU score showed a linear trajectory of improvement, starting at 6.7 after the first epoch and reaching 9.9 upon the completion of the third epoch. While constrained by the subset size, the model's progression confirms valid learning dynamics for the translation task.

V. CONCLUSION

This study validates the structural adaptation of the BART model for English-to-Turkish machine translation using a controlled dataset subset. The implemented preprocessing, tokenization, and iterative fine-tuning algorithm resulted in stable convergence and measurable improvements in translation accuracy. Future iterations of this work will aim to expand the training subset and explore more complex decoding strategies to push the BLEU metric higher.

REFERENCES

[1] M. Lewis et al., "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 7871–7880.

[2] J. Tiedemann, "Parallel Data, Tools and Interfaces in OPUS," in *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC'12)*, 2012, pp. 2214–2218.

[3] M. Post, "A Call for Clarity in Reporting BLEU Scores," in *Proceedings of the Third Conference on Machine Translation: Research Papers*, 2018, pp. 186–191.

[4] M. Acı, N. V. Sarı, and Ç. İ. Acı, "Morphological and structural complexity analysis of low-resource English-Turkish language pair using neural machine translation models," *PeerJ Computer Science*, 2025.

[5] C. Mi, L. Xie, and Y. Zhang, "Improving Adversarial Neural Machine Translation for Morphologically Rich Language," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 4, no. 4, pp. 417–426, 2020.

[6] A. Dvorkovich, S. Gubanov, and I. Galinskaya, "Yandex School of Data Analysis approach to English-Turkish translation at WMT16 News Translation Task," in *Proceedings of the First Conference on Machine Translation*, 2016, pp. 281–288.

[7] B. K. Yazar and E. Kılıç, "Improving Low-Resource Kazakh-English and Turkish-English Neural Machine Translation Using Transfer Learning and Part of Speech Tags," *IEEE Access*, 2025.

[8] B. K. Yazar, D. Ö. Şahin, and E. Kılıç, "Low-Resource Neural Machine Translation: A Systematic Literature Review," *IEEE Access*, vol. 11, pp. 131775–131813, 2023.

[9] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998–6008.

[10] R. Sennrich, B. Haddow, and A. Birch, "Neural Machine Translation of Rare Words with Subword Units," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2016, pp. 1715–1725.